

Electric Circuits Lab

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Review of KCL and Current Division Rule

April 8, 2025

Overview

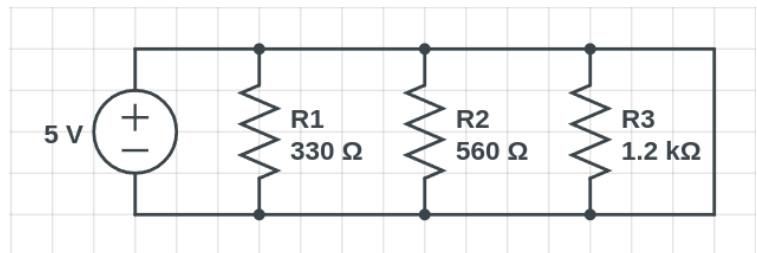
Here is book of Electric Circuits Lab experiment on page 61 (Book)

Experiment Info

- **Purpose** > To verify Kirchhoff's Current Law (KCL) and the Current Division Rule through experimental measurements.
- **Necessary Equipment** > Resistors, DC Power, Multimeter, Wires

Ohm's Law : $V = I \cdot R$

$$I_{R_1} = I_T \left(\frac{R_2 \parallel R_3}{R_1 + (R_2 \parallel R_3)} \right)$$



$$I_{R1} = \frac{5}{330} \approx 0.01515A \quad I_{R2} = \frac{5}{560} \approx 0.00893A \quad I_{R3} = \frac{5}{1200} \approx 0.00417A$$

Steps

1. First we picked resistors by their color code.
2. Then we put resistors in parallel on the breadboard.
3. Then turned off stand-by mode and measured the real resistance of each resistor.

Resistor Color and Current Table

Resistor	Color 1	Color 2	Color 3	I_{real}	$I_{measured}$
$R_1 = 330\Omega$	Orange (3)	Orange (3)	Brown (1)	15.15mA	15.32mA
$R_2 = 560\Omega$	Green (5)	Blue (6)	Brown (1)	8.93mA	8.61mA
$R_3 = 1200\Omega$	Brown (1)	Red (2)	Red (2)	4.17mA	4.32mA

Calculations of V_{real} :

$$I_{R_1} = \frac{V_1}{R_1} = \frac{5}{330} \approx 0.01515A = 15.15mA$$

$$I_{R_1} = \frac{V_1}{R_1} = \frac{5}{560} \approx 0.00893A = 8.93mA$$

$$I_{R_1} = \frac{V_1}{R_1} = \frac{5}{1200} \approx 0.00417A = 4.17mA$$

Analyzing R-L-C Circuit - Error Calculations

Error calculations are divided into 3 categories, absolute, relative and systematic errors.

from absolute formula :

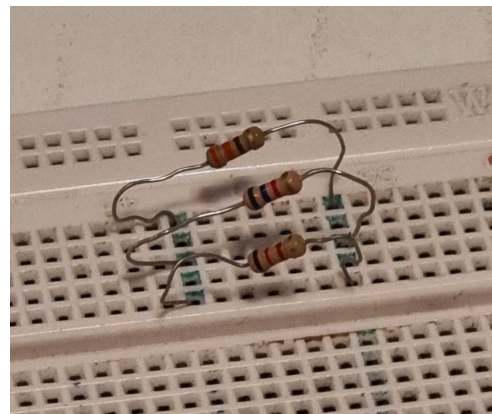
$$Absolute\ Error = |real\ value - measured\ value|$$

We get this absolute error for α :

$$\varepsilon_{abs_{R_1-real}} = |15.15mA - 15.32mA| = 0.17mA$$

$$\varepsilon_{abs_{R_2-real}} = |8.93mA - 8.61mA| = 0.32mA$$

$$\varepsilon_{abs_{R_3-real}} = |4.17mA - 4.32mA| = 0.15mA$$



from relative formula :

$$\text{Relative Error} = \frac{|real\ value - measured\ value|}{real\ value}$$

We get this absolute error for I_{1-real} :

$$\varepsilon_{rel_{I_{1-real}}} = \frac{0.17mA}{15.15mA} \approx 0.01122$$

We get this absolute error for I_{2-real} :

$$\varepsilon_{rel_{I_{2-real}}} = \frac{0.32mA}{8.93mA} \approx 0.035834$$

We get this absolute error for I_{3-real} :

$$\varepsilon_{rel_{I_{3-real}}} = \frac{0.15mA}{4.17mA} \approx 0.035971$$

for systematic formula, We must do several mathematics operations :

1. Consider we have formula $V = I \cdot R$
2. Take the $\ln$ from both sides.
3. Take the differential from both sides.
4. Convert differentials to delta (Δ).
5. Convert every $-$ to $+$.
6. accuracy of **Analog Device** is equal to the lowest shown range on the device.
7. Error of **Analog Device** is equal to the lowest shown range on device multiplied by 0.5.
8. accuracy of **Digital Device** is equal to 10^{-x} where x is float level on the screen.
9. Error of **Digital Device** is equal to 10^{-x} where x is float level on the screen.

Now we can calculate systematic error using main formula :

Step 1: $I = \frac{V}{R}$

Step 2: $\ln(I) = \ln(V) - \ln(R)$

Step 3: $\frac{dI_n}{I_n} = \frac{d(V_{total})}{V_{total}} - \frac{d(R_n)}{R_n}$

Step 4: $\frac{\Delta I_n}{I_n} = \frac{\Delta(V_{total})}{V_{total}} - \frac{\Delta(R_n)}{R_n}$

Step 6~: $\Delta I = 10^{-2}$, $\Delta R = 10^{-3}$

$$\text{for } I_1 : \varepsilon_{\text{sys}_{I_1}} = \frac{\Delta I_1}{I_1} = \frac{0.01}{15.15} - \frac{0.001}{0.330} \approx 0.0037$$

$$\text{for } I_2 : \varepsilon_{\text{sys}_{I_2}} = \frac{\Delta I_2}{I_2} = \frac{0.01}{8.93} - \frac{0.001}{0.560} \approx 0.0007$$

$$\text{for } I_3 : \varepsilon_{\text{sys}_{I_2}} = \frac{\Delta I_3}{I_3} = \frac{0.01}{4.17} - \frac{0.001}{1200} \approx 0.0024$$